Well, truth be told, in special relativity, Einstein's main focus was on a special kind of motion: constant-velocity motion. It was not until 1915, some ten years later, that he fully came to grips with more general, accelerated motion, through his general theory of relativity. Even so, Einstein and others repeatedly considered the question of rotating motion using the insights of special relativity; they concluded, like Newton and unlike Mach, that even in an otherwise completely empty universe you would feel the outward pull from spinning—Homer would feel pressed against the inner wall of a spinning bucket; the rope between the two twirling rocks would pull taut.[7] Having dismantled Newton's absolute space and absolute time, how did Einstein explain this?

The answer is surprising. Its name notwithstanding, Einstein's theory does not proclaim that everything is relative. Special relativity does claim that *some* things are relative: velocities are relative; distances across space are relative; durations of elapsed time are relative. But the theory actually introduces a grand, new, sweepingly absolute concept: *absolute spacetime.*Absolute spacetime is as absolute for special relativity as absolute space and absolute time were for Newton, and partly for this reason Einstein did not suggest or particularly like the name "relativity theory." Instead, he and other physicists suggested *invariance theory,* stressing that the theory, at its core, involves something that everyone agrees on, something that is not relative.[8]

Absolute spacetime is the vital next chapter in the story of the bucket, because, even if devoid of all material benchmarks for defining motion, the absolute spacetime of special relativity provides a something with respect to which objects can be said to accelerate.

Carving Space and Time

To see this, imagine that Marge and Lisa, seeking some quality together-time, enroll in a Burns Institute extension course on urban renewal. For their first assignment, they are asked to redesign the street and avenue layout of Springfield, subject to two requirements: first, the street/avenue grid must be configured so that the Soaring Nuclear Monument is located right at the grid's center, at 5th Street and 5th Avenue, and, second, the designs must use streets 100 meters long, and avenues, which